

9 Examples

9.1 Counting Sort Algorithm

4 6 4 1 3 4 1 4

Assume $[0 \dots k]$. Max integer: 6, size of array: 8.

Original Array: [4 | 6 | 4 | 1 | 3 | 4 | 1 | 4]

Counting Array: [0 | 2 | 0 | 2 | 3 | 0 | 1] (size is max + 1)

Update Array: [0 | 2 | 2 | 4 | 7 | 7 | 8]

Output Array: [1 | 1 | 3 | 4 | 4 | 4 | 6]

(reverse order of original array via indexes, the output is not index based)

9.2 Growth Classes Example

Show $20n^2 + 2n + 5 = O(n^2)$

$$O(n^2) = 20n^2 + 2n + 5 \quad (1)$$

$$= 20n^2 + 2n^2 + 5n^2 \quad (2)$$

$$= 27n^2 \quad (3)$$

So $n_0 = 1$ and $C = 27$.

9.3 Growth Example II

$5n + 7 = o(n^2)$. For every c , find $n_0 > 0$

$$5n + 7 \leq 5n + n = 6n < cn^2 \quad (4)$$

For $n_0 = 7$, this holds. But now we have to prove $\forall c$. We do so via limit.

9.4 Growth Class Example

Prove $5n^2 + 2n = \Theta(n^2)$. Just take big- O and big- Ω

9.5 Growth Class Example

Prove $5n^2 + 15n + 100 = \Omega(n^2)$

$$> 5n^2 - 15n \quad (5)$$

$$\geq 5n^2 - n^2 \quad (6)$$

$$= 4n^2 \quad (7)$$

9.6 Master Theorem